Splunk · Quick concepts

Interview revision sheet. Read it the evening before, then test yourself in Interview practice.

Architecture

- Universal forwarder collects and sends data; heavy forwarder can parse, filter and route.
- Indexer stores and indexes data; search head runs searches and dashboards.
- Deployment server pushes config to forwarders; cluster manager manages indexer replication.
- Ports: 8000 web, 8089 management, 9997 forwarder receiving.

Data and config

- Index = data store; sourcetype = how data is parsed; source and host describe origin.
- inputs.conf (what to collect), props.conf (parsing, timestamps), transforms.conf (routing, masking), indexes.conf (retention).
- Buckets: hot, warm, cold, frozen. frozenTimePeriodInSecs controls retention.
- btool shows the merged config; local overrides default.

Core SPL

- Start with index and sourcetype, then filter, then transform.
- stats count by host; timechart span=5m count; top and rare; table; dedup; sort - count.
- eval creates fields; rex extracts with regex; rename; fields - to drop fields.
- spath for JSON; lookup and outputlookup for enrichment.

Advanced SPL

- eventstats adds aggregates to each event; streamstats calculates running values.
- tstats is fast on indexed fields and data models.
- Avoid join and transaction when stats can do the job; subsearches have limits.
- Macros, event types, tags and the CIM normalise searches.

Alerts and dashboards

- Scheduled alerts are cheaper than real-time; throttle repeats.
- Base search plus post-process searches speed up dashboards.
- Summary indexing for long-range reports.

Troubleshooting

- index=_internal log_level=ERROR for Splunk's own errors.
- Monitoring Console for indexing rate, skipped searches and resource use.
- Wrong timestamps: check TIME_FORMAT, TIME_PREFIX and TZ in props.conf.
- Mask sensitive data at index time with SEDCMD or transforms.
- Job Inspector shows where search time goes. HEC accepts events over HTTP with tokens.